Project Initialization and Planning Phase

Date	06 July 2026
Team ID	
Project Name	Ledgerline – Credit Card Approval Prediction System
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) Report

This proposal aims to transform credit card approval decisions using machine learning, boosting consistency and speed. It tackles the inefficiency of manual review, promising faster decisions, reduced inconsistency, and clearer outcomes for applicants. Key features include a compared multi-algorithm credit-risk model and instant, calibrated decision-making.

Project Overview

Field	Description
Objective	Automate the credit card approval decision using machine learning, ensuring faster and more consistent assessments than manual review.
Scope	Covers the full pipeline from applicant data to decision: feature engineering, training and comparing 4 ML classifiers, and deploying the best one in a Flask web app supporting single-applicant, batch, and (optionally) cloud-hosted scoring.

Problem Statement

Field	Description
Description	Manually reviewing thousands of daily credit card applications against income, employment, and credit history is slow and inconsistent, and applicants get no visibility into the decision.
Impact	Solving this reduces analyst workload, speeds up decisions for genuine applicants, and gives compliance teams a consistent way to flag high-risk applicants from raw payment-history data.

Proposed Solution

Field	Description
Approach	Train and compare 4 ML classifiers on applicant financial and credit-history data; select the best by ROC-AUC; calibrate the decision threshold; serve it through a Flask web app.
Key Features	• ML-based credit-risk model (Logistic Regression, Decision Tree, Random Forest,

Field	Description
	XGBoost compared) • Instant decision-making with a confidence score • Batch CSV screening for compliance review • Model transparency report (comparison metrics, ROC curves, feature importance)

Resource Requirements

Resource Type	Description	Specification / Allocation
Computing Resources	CPU/GPU specifications	Standard CPU — no GPU required for these model sizes
Memory	RAM specifications	4–8 GB sufficient
Storage	Disk space for data, models, and logs	< 1 GB (CSV data + serialized models + charts)
Frameworks	Python frameworks	Flask
Libraries	Additional libraries	scikit-learn, XGBoost, pandas, NumPy, matplotlib, seaborn
Development Environment	IDE	Jupyter Notebook, VS Code
Data	Source, size, format	Synthetic data modeled on the Kaggle "Credit Card Approval Prediction" dataset schema (application_record.csv, credit_record.csv), 8,000 applicants, csv